general feeling that these were mere details, which, no doubt, would soon be addressed.

Within a decade, everything changed. As anticipated, the two problems Kelvin had raised were promptly addressed, but they proved anything but minor. Each ignited a revolution, and each required a fundamental rewriting of nature's laws. The classical conceptions of space, time, and reality—the ones that for hundreds of years had not only worked but also concisely expressed our intuitive sense of the world—were overthrown.

The relativity revolution, which addressed the first of Kelvin's "clouds," dates from 1905 and 1915, when Albert Einstein completed his special and general theories of relativity (Chapter 3). While struggling with puzzles involving electricity, magnetism, and light's motion, Einstein realized that Newton's conception of space and time, the cornerstone of classical physics, was flawed. Over the course of a few intense weeks in the spring of 1905, he determined that space and time are not independent and absolute, as Newton had thought, but are enmeshed and relative in a manner that flies in the face of common experience. Some ten years later, Einstein hammered a final nail in the Newtonian coffin by rewriting the laws of gravitational physics. This time, not only did Einstein show that space and time are part of a unified whole, he also showed that by warping and curving they participate in cosmic evolution. Far from being the rigid, unchanging structures envisioned by Newton, space and time in Einstein's reworking are flexible and dynamic.

The two theories of relativity are among humankind's most precious achievements, and with them Einstein toppled Newton's conception of reality. Even though Newtonian physics seemed to capture mathematically much of what we experience physically, the reality it describes turns out not to be the reality of our world. Ours is a relativistic reality. Yet, because the deviation between classical and relativistic reality is manifest only under extreme conditions (such as extremes of speed and gravity), Newtonian physics still provides an approximation that proves extremely accurate and useful in many circumstances. But utility and reality are very different standards. As we will see, features of